

Class 13: RNASeq Analysis

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Background

Today we will analyze some RNA sequencing data on the effects of a common steroid drug on airway cell lines.

There are two main inputs we need for their analysis:

- countData: counts for genes in rows with experiments in the columns
- colData: or metadata

Import countData and colData

```
counts <- read.csv("airway_scaledcounts.csv", row.names=1)
metadata <- read.csv("airway_metadata.csv")
```

Q1. How many genes are in this dataset?

```
nrow(counts)
```

```
[1] 38694
```

Q2. How many ‘control’ cell lines do we have?

```
sum(metadata$dex == "control")
```

```
[1] 4
```

Toy Differential Gene Expression

Let’s try finding the average or mean of the “control” and “treated” columns and see if they differ.

- First we need to find all “control” columns
- Extract just the “control” values for each gene
- Calculate the `mean()` for each gene “control” values

```
metadata
```

	id	dex	celltype	geo_id
1	SRR1039508	control	N61311	GSM1275862
2	SRR1039509	treated	N61311	GSM1275863
3	SRR1039512	control	N052611	GSM1275866
4	SRR1039513	treated	N052611	GSM1275867
5	SRR1039516	control	N080611	GSM1275870
6	SRR1039517	treated	N080611	GSM1275871
7	SRR1039520	control	N061011	GSM1275874
8	SRR1039521	treated	N061011	GSM1275875

```
all(colnames(counts) == metadata$id)
```

```
[1] TRUE
```

the dex column tells me whether we have “control” or “treated”

```
control.inds <- metadata$dex == "control"
```

Extract just the “control” values for all genes

```
control.counts <- counts[,control.inds]
```

Calculate the mean value for each gene in these control columns

```
control.mean <- rowMeans( control.counts )  
head(control.mean)
```

```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460  
          900.75           0.00           520.50           339.75           97.25  
ENSG000000000938  
          0.75
```

Q3. Do the same for the “treated” values to get a `treated.mean`

```
treated.inds <- metadata$dex == "treated"  
treated.counts <- counts[,treated.inds]  
treated.mean <- rowMeans( treated.counts )  
head(treated.mean)
```

```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460  
          658.00           0.00           546.00           316.50           78.75  
ENSG000000000938  
          0.00
```

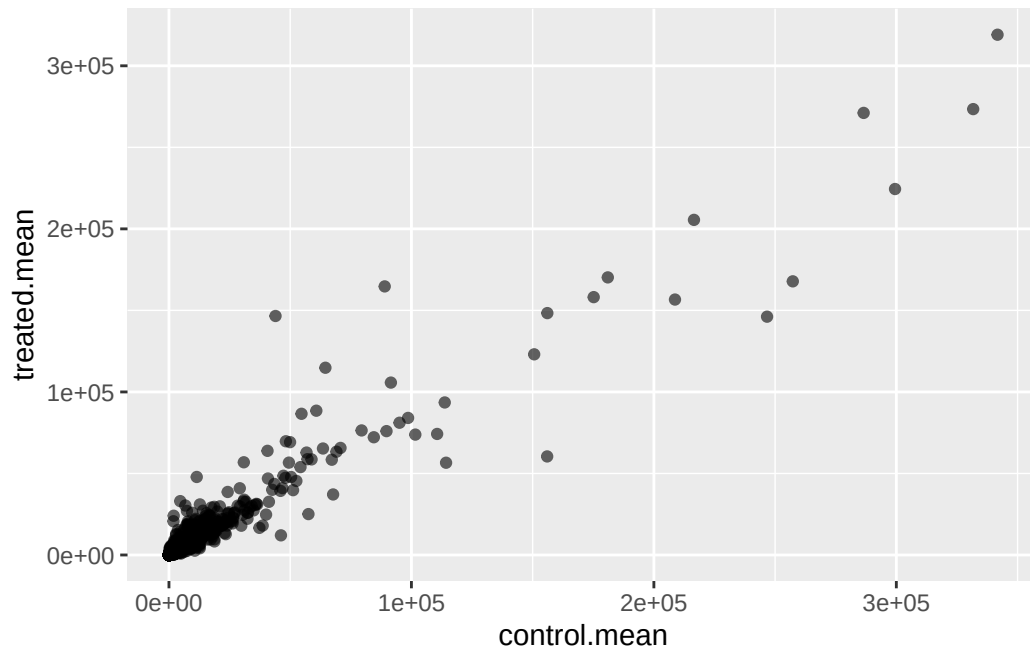
Q4. Make a plot of `control.mean` vs `treated.mean`

Lets store our mean values together in a data.frame for easier book-keeping.

```
meancounts <- data.frame(control.mean, treated.mean)  
head(meancounts)
```

	control.mean	treated.mean
ENSG000000000003	900.75	658.00
ENSG000000000005	0.00	0.00
ENSG000000000419	520.50	546.00
ENSG000000000457	339.75	316.50
ENSG000000000460	97.25	78.75
ENSG000000000938	0.75	0.00

```
library(ggplot2)
ggplot(meancounts) +
  aes(control.mean, treated.mean) +
  geom_point(alpha=.6)
```

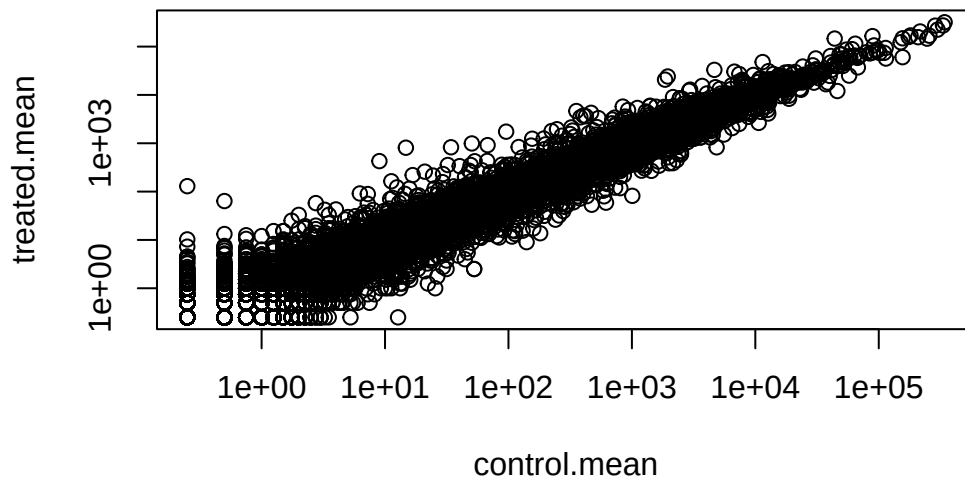


We totally need to log transform this data as it is heavily skewed!

```
plot(meancounts, log="xy")
```

Warning in xy.coords(x, y, xlabel, ylabel, log): 15032 x values <= 0 omitted from logarithmic plot

Warning in xy.coords(x, y, xlabel, ylabel, log): 15281 y values <= 0 omitted from logarithmic plot



Amount is the same

```
log(20/20)
```

```
[1] 0
```

Doubling the amount

```
log(40/20)
```

```
[1] 0.6931472
```

Halving the amount

```
log(10/20)
```

```
[1] -0.6931472
```

A common rule of thumb is to focus on genes with a log₂ “fold change” of +2 as so called UP REGULATED and -2 as DOWN REGULATED.

```
log2(80/20)
```

```
[1] 2
```

Lets add a log2 fold change value to our `meancounts` dataframe.

```
meancounts$log2fc <- log2(meancounts$treated.mean/meancounts$control.mean)
head(meancounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000005	0.00	0.00	NaN
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805
ENSG000000000460	97.25	78.75	-0.30441833
ENSG000000000938	0.75	0.00	-Inf

Q. Remove any “zero count” genes from the dataset for further analysis

```
to.keep <- rowSums(meancounts[,1:2] == 0) == 0
sum(to.keep)
```

```
[1] 21817
```

```
mycounts <- meancounts[to.keep,]
head(mycounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805
ENSG000000000460	97.25	78.75	-0.30441833
ENSG000000000971	5219.00	6687.50	0.35769358
ENSG000000001036	2327.00	1785.75	-0.38194109

Q. How many genes are “up” regulated a a log2fc threshold of +2?

```
sum(mycounts$log2fc >= 2)
```

[1] 314

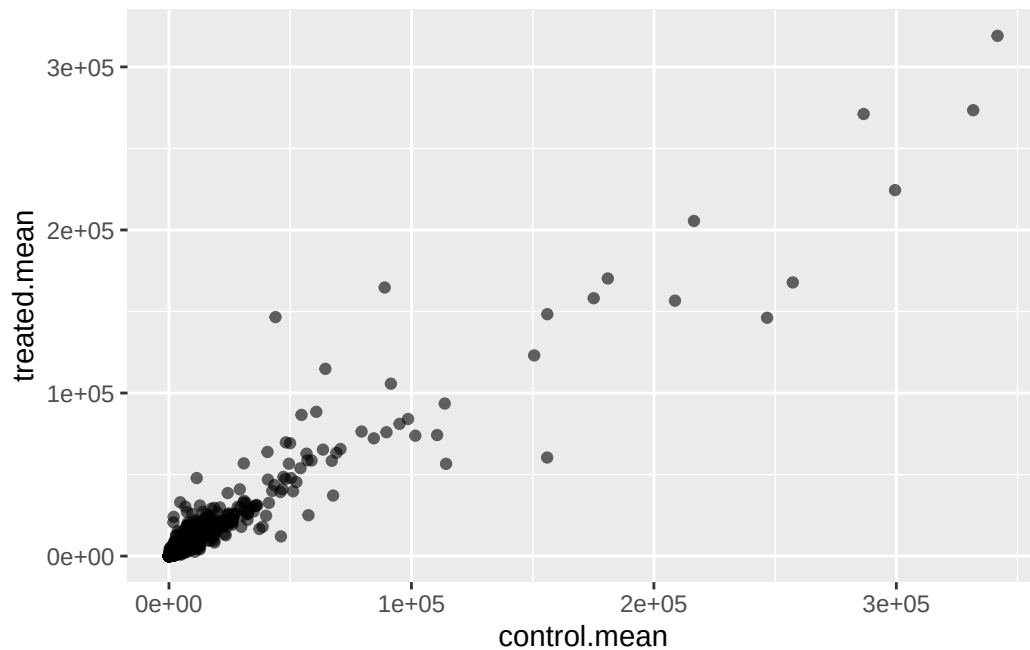
Q. How many genes are “down” regulated a a log2fc threshold of -2?

```
sum(mycounts$log2fc <= -2)
```

[1] 485

However, there are some stats missing from this analysis, such as how significant these changes are and/or whether differences between the control and treated may have other causes.

```
ggplot(mycounts) +  
  aes(control.mean, treated.mean) +  
  geom_point(alpha=.6)
```



Setting up for DESeq

We will use DESeq2 to do this:

```
library(DESeq2)
```

The first function we will use from this package sets up the input in the particular format that DESeq wants.

```
dds <- DESeqDataSetFromMatrix(countData = counts,  
                              colData = metadata,  
                              design = ~dex)
```

converting counts to integer mode

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

We can now run our DESeq analysis

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
res <- results(dds)
```

Peek at results

```
head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

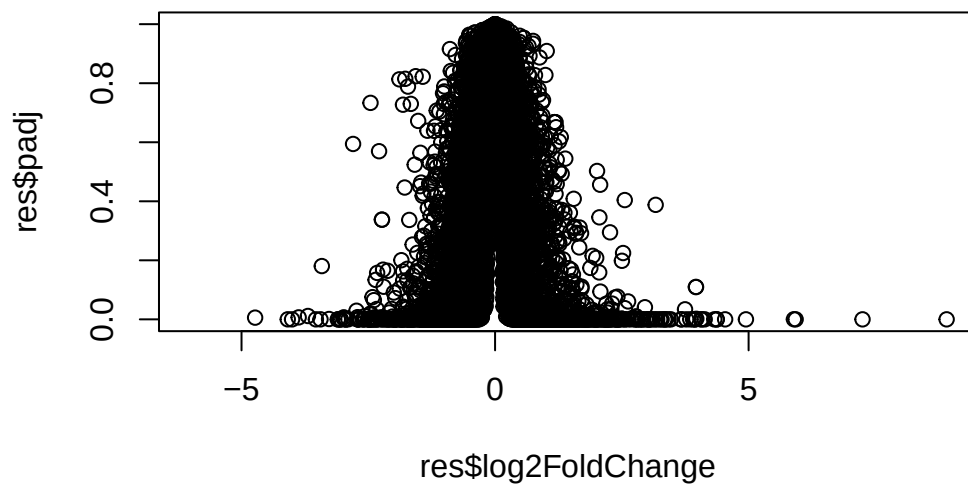
DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.3507030	0.168246	-2.084470	0.0371175
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG0000000000419	520.134160	0.2061078	0.101059	2.039475	0.0414026
ENSG0000000000457	322.664844	0.0245269	0.145145	0.168982	0.8658106
ENSG0000000000460	87.682625	-0.1471420	0.257007	-0.572521	0.5669691
ENSG0000000000938	0.319167	-1.7322890	3.493601	-0.495846	0.6200029
	padj				
	<numeric>				
ENSG0000000000003	0.163035				
ENSG0000000000005	NA				
ENSG0000000000419	0.176032				
ENSG0000000000457	0.961694				
ENSG0000000000460	0.815849				
ENSG0000000000938	NA				

Result figure: Volcano Plot

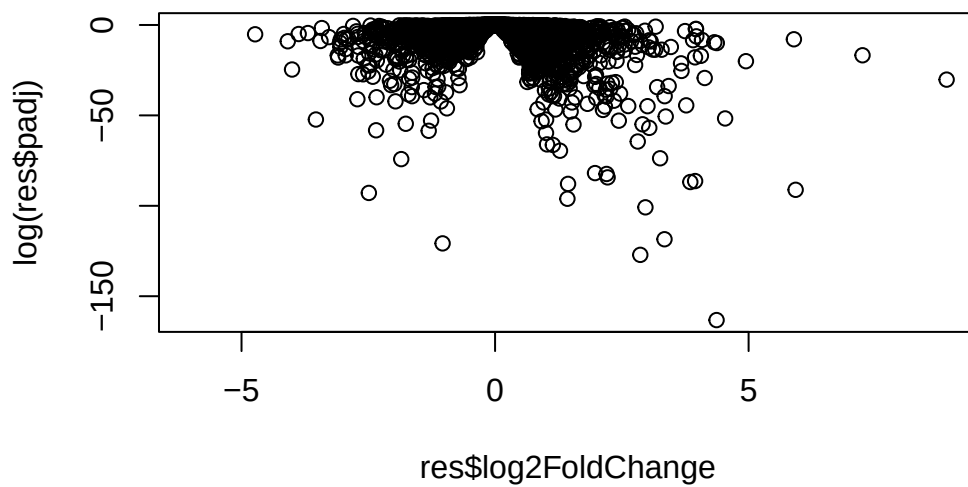
plot of the P values the Log2FC

```
plot(res$log2FoldChange, res$padj)
```



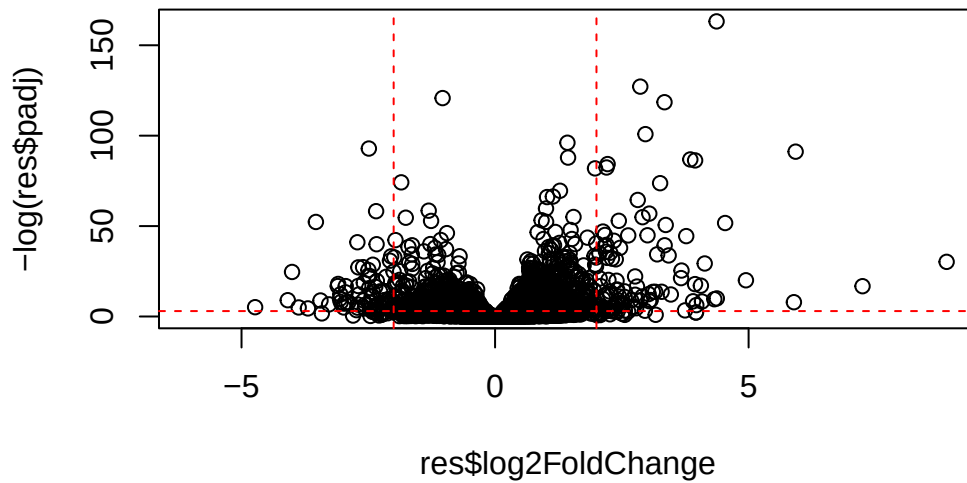
This P-value is again heavily skewed so lets log transform it.

```
plot(res$log2FoldChange, log(res$padj))
```

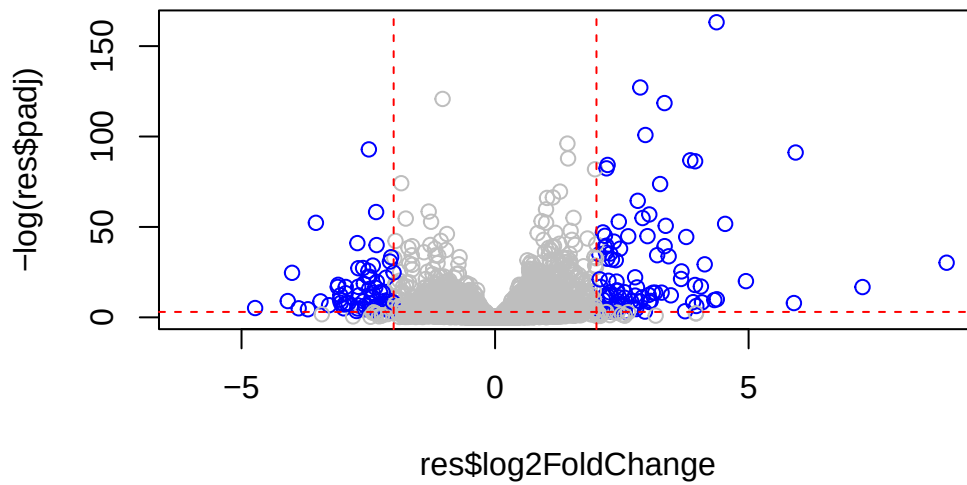


We can flip the y-axis by adding a minus sign. This will make it easier to interpret.

```
plot(res$log2FoldChange, -log(res$padj))  
# Add some cut-off lines  
abline(v=c(-2,2), col="red", lty=2)  
abline(h=-log(0.05), col="red", lty=2)
```



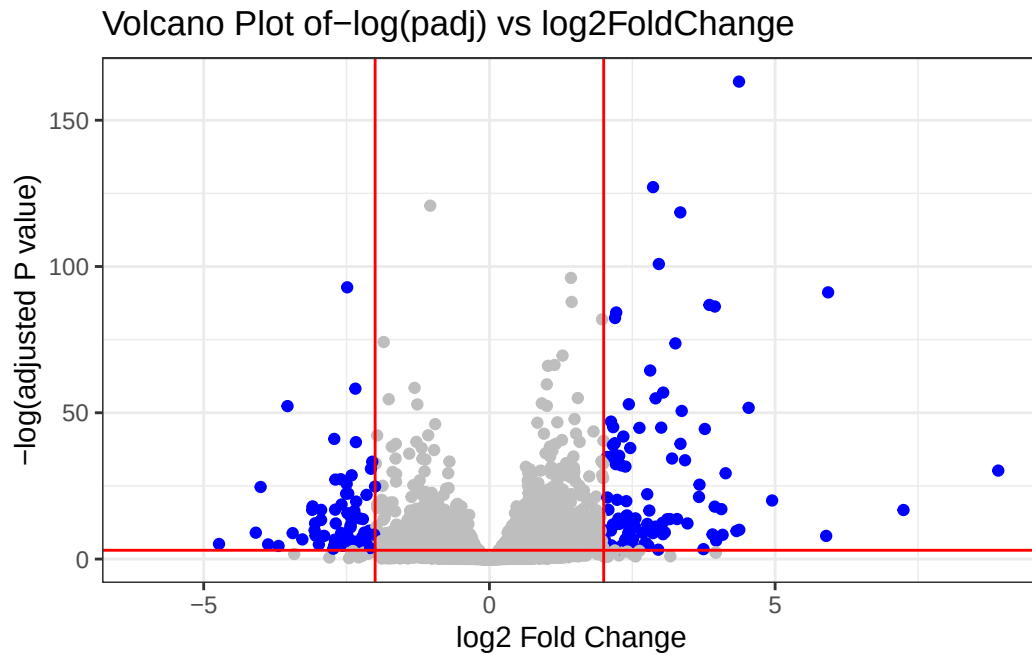
```
mycols <- rep("grey", (nrow(res)))  
mycols[ abs(res$log2FoldChange) >= 2 ] <- "blue"  
mycols[ res$padj > 0.05 ] <- "grey"  
plot(res$log2FoldChange, -log(res$padj), col=mycols)  
  
abline(v=c(-2,2), col="red", lty=2)  
abline(h=-log(0.05), col="red", lty=2)
```



Q. Make a ggplot volcano plot with colors and lines as annotation along with nice axis labels.

```
ggplot(as.data.frame(res)) +
  aes(x = log2FoldChange, y = -log(padj)) +
  geom_point(col = mycols) +
  labs(title = "Volcano Plot of -log(padj) vs log2FoldChange",
       x = "log2 Fold Change",
       y = "-log(adjusted P value)") +
  geom_vline(xintercept = c(-2, 2), color = "red") +
  geom_hline(yintercept = -log(0.05), color = "red") +
  theme_bw()
```

Warning: Removed 23549 rows containing missing values or values outside the scale range (`geom\_point()`).



We need to add gene annotation

Gene symbols and different database IDs

Pathway Analysis

Find what biological psthways my differentially expressed genes participate in.